Page1 UNIT-1

Concept of manufacturing process

Manufacturing processes can also be classified based on the type of material being used, such as metals, plastics, ceramics, and composites. Additionally, they can be categorized based on the level of automation involved, from manual processes to fully automated systems. Other factors that can impact the classification of manufacturing processes include the level of precision required, the complexity of the product, and the scale of production.

Metal forming process

Metal forming is a process where materials are subjected to plastic deformation to obtain the required size, shape, and/or change the physical and chemical properties. Metal forming is divided into two groups, bulk forming, and sheet forming.

Metal machining process

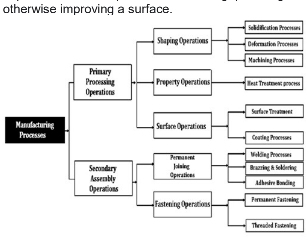
The controlled process of cutting a piece of raw material and creating a desired final shape and size is known as machining. Metal machining uses milling machines, lathes, drill presses and various other machines to manufacture shapes in a wide variety of metals.

Metal joining process

Metal joining is a process that uses heat to melt or heat metal just below the melting temperature. Joining metal by fusion is known as fusion welding. Without fusion, the process is known as solid-state welding.

Metal finishing processes

Metal finishing is an all-encompassing term used to describe the process of placing some type of metal coating on the surface of a metallic part, typically referred to as a substrate. It can also involve the implementation of a process for cleaning, polishing or otherwise improving a surface.



UNIT-2

Hot working-Hot working is plastically deforming of the metallic material at a temperature above the recrystallization temperature. Example: Hot rolling, Hot forging, Hot extrusion etc

ADVANTAGES-

•Bulk deformation can be achieved, so bulk dimensional and shape change can be done

. •Very negligible change in mechanical properties during working.

•Ductility of material is retained.

•No strengthening occurs during hot working.

DISADVANTAGES-

•No dimentional accuracy can be achived.

•Rough surfaces created.

•Metallic oxide forms on the surface of workpiece during processing.

COLD WORKING-Cold working may be defined as plastic deformation of metals and alloys at a temperature below the recrystallisation temperature for that metal or alloy. Example: Cold rolling, cold forging, cold extrusion etc

ADVANTAGES

•Good dimensional control

•Good surface finish of the component.

•Strength and hardness of the metal are increased.

•An ideal method for increasing hardness of those metals which do not respond to the heat treatment-

PRINCIPLE OF FORGING PROCESS-Forging is a manufacturing process involving the shaping of a metal through hammering, pressing, or rolling. These compressive forces are delivered with a hammer or die. Forging is often categorized according to the temperature at which it is performed—cold, warm, or hot forging. A wide range of metals can be forged.

Forging temperature and grain flow in forged parts

The forging temperatures are above the recrystallization temperature, and are typically between 950°C–1250°C. Usually, one experiences good formability (i.e., filling of die-cavity in the context of forging), low forming forces, and an almost uniform tensile strength of the work-piece

Classification of forging processes

1.DROP FORGING
I)BOARD DROP HAMMER
II)BELT DROP HAMMER
III)POWER DROP HAMMER
2.PRESS FORGING
I)MECHANICAL PRESS
II)HYDRAULIC PRESS
III)SCREW PRESS
3.OPEN DIE FORGING
4.CLOSED DIE FORGING

OPEN DIE

In open-die forging, a hammer strikes and deforms the workpiece, which is placed on a stationary anvil. Open-die forging gets its name from the fact that the dies (the surfaces that are in contact with the workpiece) do not enclose the workpiece, allowing it to flow except where contacted by the dies. Therefore the operator needs to orient and position the workpiece to get the desired shapeOpen die forging is an important technique for many types of manufacturing. It allows rough and finishing shaping of metal, most commonly steel and steel alloys.

Closed die or impression die forging

Closed Die Forging is a forging process in which dies (called tooling) that contain a precut profile of the desired part move towards each other and covers the workpiece in whole or in part. The heated raw material, which is approximately the shape or size of the final forged part, is placed in the bottom die. The shape of the forging is incorporated in the top or bottom die as a negative image. Coming from above, the impact of the top die on the raw material forms it into the required forged form.

As the name implies, two or more dies containing impressions of the part shape are brought together as forging stock undergoes plastic deformation. Because metal flow is restricted by the die contours, this process can yield more complex shapes and closer tolerances than open-die forging processes. Additional flexibility in forming both symmetrical and non-symmetrical shapes comes from various pre-forming operations (sometimes bending) prior to forging in finisher dies.

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BOARD DROP HAMMER

The board drop hammer is a drop forging machine tool that relies only on gravity. A hardwood board is attached to the ram, rollers grip the board and can raise the board and ram due to friction forces between the board and rollers. Once the ram is raised to the height needed, the rollers can be pulled apart and the apparatus will be released, sending the forging hammer on its way.

BELT DROP HAMMER

The belt drop hammer is another drop forging machine tool that operates similarly to the board drop hammer. Rollers grip a belt that is connected to the ram. They raise the hammer by the belt, this causes the belt to gain slack. Pulling apart the rollers will release the belt. When the belt is released, the hammer will fall. This type of machine also relies only on the force of gravity.

POWER DROP HAMMER

Power drop hammers use the force of pressurized air or steam to raise the ram, and to accelerate it downward to strike the work. Power drop hammers can accelerate the ram downward to a higher speed than conventional drop forging machine tools, they can also accommodate a more massive ram. Power drop hammers can deliver much more energy per blow than conventional drop forging hammers.

Significance of grain flow direction: Mechanical properties vary with respect to orientation relative to grain flow. Properties like impact toughness, ductility and fatigue strength, which are measures of a material's resistance to cracking, can be improved significantly by proper alignment of the direction of crack propagation and the grain flow lines. The optimum alignment occurs when the maximum principal stress (perpendicular to a potential crack or fracture) is aligned with the grain-flow lines.

Forging temperature is the temperature at which a metal becomes substantially more soft, but is lower than the melting temperature. Bringing a metal to its forging temperature allows the metal's shape to be changed by applying a relatively small force, without creating cracks. The forging temperature of an alloy will lie between the temperatures of its component metals. For most metals, forging temperature will be approximately 70% of the melting temperature in kelvins.

• Edging" is the process of concentrating material using a concave shaped open-die. The process is called "edging" because it is usually carried out on the ends of the workpiece. which the metal piece is displaced to the desired shape by striking between two dies edging is frequently as primary drop forging operation

Forging defects

Forging is a manufacturing process in which metal is shaped by applying compressive forces. Although forging is generally considered to be a reliable and high-quality manufacturing process, it is still subject to certain defects that can arise during the process. Some common forging defects include:

Cracks: Cracks can occur when the metal is subjected to too much stress during forging. This can happen if the temperature is too low or if the forging process is done too quickly.

Cold shuts: Cold shuts occur when two portions of metal fail to weld together properly during the forging process. This can be caused by poor quality metal or by insufficient heat.

Inclusions: Inclusions are impurities that are trapped within the metal during the forging process. They can cause weakness in the final product and can lead to cracks or other defects.

Misruns: Misruns occur when the metal fails to fill the entire die cavity during the forging process. This can be caused by poor die design or insufficient pressure.

Surface irregularities: Surface irregularities, such as pits or scratches, can occur during the forging process due to poor die design or improper handling of the metal.

Laps: Laps occur when a portion of the metal folds back on itself during the forging process. This can be caused by insufficient pressure or poor die design.

Rolling and Extrusion:

PRINCIPLE OF ROLLING : The rolling process is a metal forming process, in which stock of the material is passed between one or more pairs of rollers in order to reduce and to maintain the uniform thickness. The metal is compressed by the rollers while simultaneously shifting it forward due to the friction at the roller interfaces. As soon as the workpiece passes through rollers, the operation is fully done. That is, the thickness of the work decreases while the length and width increases. the increment in the width is insignificant and always neglected. The increase in the length is known as absolute elongation, while an increase in the width s known as an absolute spread. Also, the decrease in the thicknessis called a draft.

PRINCIPLE OF EXTRUSION :

Metal Extrusion is a metal forming manufacturing process in which a cylindrical billet inside a closed cavity is forced to flow through a die of a desired crosssection. These fixed cross-sectional profile extruded parts are called "Extrudates" and pushed out using either a mechanical or hydraulic press. Most commonly extruded materials are Aluminium, Copper, Steel, Magnesium, and Lead.

Hot rolling: When rolling temperature is above recrystallisation temperature of work material, then the process is called hot rolling.

Cold rolling: When rolling temperature is below recrystallisation temperature of work material, then the process is called cold rolling.

Two-High Rolling Mills

Two High Reversing Mill: In this type of mill, the rollers are both adjustable. In these mills, rotation of that two rolls is made in two different directions. In this operation, the metal is passed between two rollers that rotate at the same speed but it is in the opposite direction. It is used in slabbing, plumbing, rail, plate roughing work and many other areas. As there is the need for a reversible drive, this mill is cheaper compared to the others.

Two High Non-Reversing Mills: In these mills, two rolls continuously revolve in the same direction and we can't reverse the direction of the rollers. In this operation, the motive power is less costly.

Three-High Rolling Mills In this mill, the three rolls stand in parallel one by others. The rolls are rotating in opposite directions. In this mill, between the first and the second rolls, the material passes. If the second roll rotates in a direction then the bottom roll rotates in another direction. The material is rolled both in forward and return in three high rolling mills. At first, it passes forward through the last and second roller and then comes back through the first and second roller. In that mill, the thickness of the material is reduced and being uniform by each pass. Here transition system and a are needed which is less powerful.

Four High Rolling Mills In this type of mill, there are four parallel rolls one by another. In this operation, the rotation of the first and the fourth rolls take place in the opposite direction of the second and the third rolls. The second and third rolls are smaller to provide rigidity in necessity. So those are known as back up rolls. It is used in the hot rolling process of the armor and in the cold rolling process of sheets, strips, and plates.

ROLLING DEFECTS :

Edge Cracks: These are cracks that appear on the edges of the rolled metal due to high stress concentrations.

Alligatoring: This is a type of defect where the metal surface becomes covered with a series of parallel cracks that resemble the scales on an alligator's skin.

Center Cracks: These are cracks that appear in the center of the rolled metal due to uneven cooling or improper rolling.

Wavy Edges: This is a defect where the edges of the rolled metal become wavy, usually due to uneven rolling pressure.

Surface Imperfections: These can include scratches, pits, and dents on the surface of the metal, which can occur due to a variety of causes, such as debris or equipment malfunction.

Hot extrusion is a metalworking process used to shape materials that are too difficult to shape using conventional methods. In this process, a material is forced through a die at a high temperature and pressure, resulting in a desired shape.

Cold extrusion is a metalworking process that is similar to hot extrusion, but it is performed at or near room temperature, as opposed to at a high temperature. It is used to shape materials that are too brittle or hard to be shaped using conventional methods.

Direct Extrusion, sometimes called Forward Extrusion is the most common type of extrusion. The process as shown in figure 2 below, begins by loading a heated billet (only for hot extrusion, discussed later) into a press cavity container where a dummy block is placed behind it. Then the mechanical or hydraulic ram presses on the material to push it out through the die. Then, while still hot, the part is stretched to straighten Under direct extrusion, the high friction caused by steels at higher temperatures is reduced using molten glass as a lubricant while oils with graphite powder are used for lubrication for low temperatures. The dummy block is used to protect the tip of the pressing stem (punch or ram) in hot extrusion. When the punch reaches the end of its stroke, a small portion of the billet called "butt end" cannot be pushed through the die opening.

In Indirect Extrusion, the die is located at the end of the hydraulic ram and moves towards the billet inside the cavity to push the material through the die. This process consumes less power due to the static billet container causing less friction on the billet. However, supporting the extruded part is difficult when the extrudate exits the die.

Backward extrusion:

Backward extrusion is a metal forming process used to create cylindrical parts with a reduced diameter on one end. In this process, a cylindrical billet or workpiece is placed in a die and a punch is used to apply force to the opposite end of the billet, forcing it to flow backwards through the die. As the material flows through the die, it is constrained by the walls of the die, causing it to take on the shape of the die cavity.

Impact extrusion:

Impact extrusion is a metal forming process used to produce hollow parts with thin walls and a high degree of dimensional accuracy. In this process, a metal slug or disc is placed in a die and struck with a punch at a high speed and force, causing the metal to flow radially outward and take on the shape of the die cavity.

ROLLING

ADVANTAGES

Some of the advantages of rolling include:

Improved mechanical properties: Rolling can improve the mechanical properties of a metal workpiece by reducing its grain size, aligning its crystal structure, and increasing its strength, toughness, and ductility.

Cost-effective: Rolling is a relatively low-cost process compared to other metal forming methods, such as forging or casting, due to its high production rates and minimal material waste.

High dimensional accuracy: Rolling can produce metal products with very precise dimensions and tolerances, making it suitable for applications where tight tolerances are required.

Improved surface finish: Rolling can improve the surface finish of a metal workpiece, producing a smooth and uniform surface that is free of defects such as scratches, pits, or dents.

DISADVANTAGES

Rolling a vehicle: Increased risk of accidents, particularly if the driver is inexperienced or reckless.

Rolling a debt:Accumulation of interest charges, which can make the debt more expensive to pay off over time.

Rolling a joint: Potential legal consequences, depending on the jurisdiction and circumstances.

Rolling a task: Potential for errors or mistakes if the task is rushed or not properly planned.

APPLICATIONS:

Manufacturing: Rolling is commonly used in manufacturing processes such as metalworking, plastic processing, and papermaking. It is used to create flat or curved sheets of material, such as metal sheets, by passing them through a series of rollers.

Transportation: Rolling is a key aspect of transportation, especially in the form of wheels and tires. The rolling motion allows vehicles to move smoothly and efficiently on a surface. Rolling is also used in the design of bearings, which are used in machines to reduce friction between moving parts.

Sports: Rolling is an important aspect of many sports, such as bowling, curling, and bocce ball. In these sports, players roll a ball towards a target, using precision and strategy to score points.

Physics: Rolling motion is an important concept in physics, particularly in the study of mechanics. It is used to describe the motion of objects that rotate around an axis, such as a rolling ball or a spinning top.

EXTRUSION:

ADVANTAGES

High Efficiency: Extrusion is a highly efficient manufacturing process. It allows for the creation of complex shapes and products quickly and with minimal waste.

Versatility: Extrusion can be used to manufacture a wide range of products including tubing, pipes, seals, rods, profiles, and more. Extruded products can be made from a variety of materials including metals, plastics, and rubber.

Cost-Effective: Extrusion is a cost-effective manufacturing process as it requires minimal tooling and can produce large quantities of products in a short amount of time.

Consistency: Extruded products have consistent dimensions and high accuracy. The extrusion process can produce products with tight tolerances and uniform wall thickness.

DISADVANTAGES:

Limited shapes: Extrusion is best suited for creating objects with a constant cross-section. It is difficult to extrude complex shapes or objects with internal cavities.

Material limitations: Some materials may not be suitable for extrusion, either due to their properties or their cost. For example, some high-performance alloys may be difficult to extrude, and some plastics may not be cost-effective to extrude.

Equipment cost: Extrusion equipment can be expensive to purchase and maintain, which may make it difficult for small businesses or individuals to use the process.

Quality control: The quality of the extruded product may be affected by a number of factors, including the quality of the raw materials, the temperature of the extrusion process, and the quality of the equipment.

APPLICATIONS

Plastic extrusion: This is one of the most common uses of extrusion, where plastic is melted and extruded through a die to create products such as pipes, tubing, sheets, films, and profiles.

Food processing: Extrusion is used in the food industry to create products such as breakfast cereals, snacks, pasta, and pet food. The extrusion process can modify the texture, taste, and nutritional value of food products.

Metal extrusion: This process is used to create metal products with a consistent cross-section, such as rods, tubes, and wire. Metal extrusion is used in the aerospace, automotive, and construction industries, among others.

Pharmaceutical industry: Extrusion is used in the pharmaceutical industry to create products such as tablets, capsules, and granules. The extrusion process can help control the release of active ingredients in medication.

Press Working:

BASIC PRINCIPLES OF SHEET METAL

Press work on sheet metal is a manufacturing process that involves the use of a press machine to shape or form a flat sheet of metal into a desired shape. The basic principle of press work on sheet metal is to apply force on the sheet metal to deform it into a specific shape or profile.

The process typically involves placing the sheet metal between two dies, which are usually made of hardened steel. The top die is usually attached to a ram or piston that applies the force to the sheet metal, while the bottom die is typically fixed to a base or bed.

When the press machine is activated, the top die moves downward and applies force to the sheet metal, causing it to deform and take on the shape of the die. The amount of force applied and the duration of the press determine the degree of deformation and the final shape of the sheet metal.

Press work on sheet metal can be used to create a wide range of shapes and profiles, including bends, folds, flanges, and embossments. It is commonly used in the manufacturing of metal components for various industries, such as automotive, aerospace, and construction.

Cogging is a process of bringing the initial forging stock to proper dimention, mainly thickness and width. It is also called drawing out.

FULLERING: The rounded nose of the fuller spreads the metal more efficiently than the flat face of the hammer. The process leaves ridges in the stock, which may then be flattened out later with the hammer or other tools. "Fullering," more generally, refers to any forging process creating a sharp transition in cross-dimensional area; with care, some types of fullering can be achieved using only hammer and the edge of the anvil (which acts as the fuller)

Bending is very common forging operation. It is an operation to give a turn to metal rod or plate. This is required for those which have bends shapes..

BLOCKING: A forging operation often used to impart an intermediate shape to a forging, preparatory to forging of the final shape in the finishing impression of the dies. Blocking can assure proper "working" of the material and contribute to greater die life.

FINISHING:. Finishing is last operation done on a pre-forged material to produce the finished shape and dimention of the product.

UNIT-3

Basic concept of machining:

Machining is a manufacturing process in which a cutting tool removes material from a workpiece to shape it into a desired form. The basic principles of machining involve the following:

Cutting tool selection: The selection of the cutting tool depends on the material to be machined, the shape and size of the workpiece, and the required surface finish. The cutting tool should be hard enough to cut the workpiece material and durable enough to withstand the cutting forces.

Workpiece holding: The workpiece must be securely held in place during machining to prevent movement and ensure accurate cutting. This is typically accomplished with clamps, vices, or fixtures that are designed to hold the workpiece in the desired position.

Cutting parameters: The cutting parameters include the speed, feed rate, and depth of cut of the cutting tool. These parameters are chosen based on the material being machined and the desired surface finish. Higher cutting speeds can result in faster machining, but may also lead to increased tool wear and shorter tool life.

Coolant usage: Coolant is often used during machining to reduce the heat generated by cutting and to remove chips from the cutting area. The type of coolant used depends on the material being machined and the type of cutting tool being used.

Quality control: Machining operations require precise control to ensure that the finished part meets the required specifications. Quality control measures may include monitoring the cutting parameters, inspecting the workpiece during machining, and using measuring tools to check the dimensions of the finished part.

Basic concept of cutting speed:

Cutting speed refers to the speed at which the cutting tool moves across the workpiece during a machining operation. It is usually expressed in surface feet per minute (SFPM) or meters per minute (m/min).

The cutting speed is determined by the material being machined, the type of cutting tool being used, and the desired surface finish. The cutting speed is directly proportional to the diameter of the cutting tool, meaning that a larger diameter tool can be used at a higher cutting speed than a smaller diameter tool.

In general, a higher cutting speed results in a higher material removal rate, which means that more material can be removed in a given amount of time. However, a higher cutting speed can also result in a shorter tool life and a rougher surface finish, so it is important to balance cutting speed with other machining parameters such as feed rate and depth of cut.

Cutting speed can be calculated using the following formula:

Cutting speed (SFPM) = (pi x tool diameter x RPM) / 12

where pi is the mathematical constant, tool diameter is the diameter of the cutting tool in inches, RPM is the spindle speed in revolutions per minute, and 12 is a conversion factor to convert from inches to feet.

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Basic concept of Feed: Feed in machining refers to the rate at which the cutting tool is moved relative to the workpiece during a machining operation. It is one of the important parameters in determining the efficiency and quality of the machining process. Here are some basic concepts related to feed in machining:

Feed rate: The feed rate is the distance the cutting tool moves along the workpiece in a given amount of time. It is usually measured in units of inches per minute (IPM) or millimeters per minute (mm/min).

Cutting speed: The cutting speed is the speed at which the cutting tool rotates or moves through the workpiece. It is usually measured in surface feet per minute (SFM) or meters per minute (m/min).

Depth of cut: The depth of cut is the distance the cutting tool is moved into the workpiece during each pass. It is usually measured in inches or millimeters.

Feed per tooth: The feed per tooth is the distance the cutting tool moves along the workpiece during each tooth engagement. It is usually measured in inches or milimeters.

Basic concept of depth of cut :

Depth of cut is a fundamental concept in machining, referring to the amount of material that is removed from a workpiece in a single pass by a cutting tool. The depth of cut is determined by the distance between the tool tip and the surface of the workpiece, and it is usually measured in inches or millimeters.

There are two main types of depth of cut: axial depth of cut and radial depth of cut. Axial depth of cut refers to the distance that the cutting tool penetrates into the workpiece along the axis of rotation, while radial depth of cut refers to the distance that the cutting tool moves across the surface of the workpiece.

In general, the depth of cut is determined by several factors, including the properties of the material being machined, the cutting tool geometry, and the cutting parameters, such as cutting speed, feed rate, and spindle speed. A deeper depth of cut can result in faster material removal rates but can also increase cutting forces and heat generation, leading to potential tool wear and damage.

Chip formation mechanism:.

Chip formation refers to the process of material removal during machining operations such as turning, drilling, milling, and grinding. The mechanism of chip formation is a complex process that depends on several factors, including the properties of the workpiece material, the cutting tool geometry, and the cutting conditions.

tThe primary mechanism of chip formation is plastic deformation, which occurs when the cutting tool exerts a force on the workpiece material. As the tool advances into the workpiece, it deforms the material ahead of the cutting edge, causing it to yield and flow in the direction of the cutting force. This deformation leads to the formation of a chip, which is essentially a piece of the workpiece material that has been removed by the cutting tool.

The shape and size of the chip depend on the cutting conditions, such as the cutting speed, feed rate, and depth of cut. At low cutting speeds, the chip tends to be continuous and long, while at high cutting speeds, it becomes shorter and more discontinuous. The type of chip produced can also vary depending on the workpiece material and the cutting tool geometry. For example, ductile materials such as aluminum tend to produce continuous chips, while brittle materials such as cast iron tend to produce shorter, segmented chips.

In addition to plastic deformation, other factors that can affect chip formation include heat generation and friction between the cutting tool and the workpiece. Heat generated during machining can soften the workpiece material, making it easier to deform and form a chip. However, excessive heat can also cause the cutting tool to wear more quickly, leading to poor surface finish and shorter tool life. Similarly, friction between the tool and the workpiece can affect chip formation by affecting the temperature and pressure distribution at the cutting edge.

Types of Chips: There are several types of chips that can be produced depending on the machining process and the characteristics of the workpiece material. Here are some common types of chips in machining:

Continuous chips: Continuous chips are long and narrow, and they are produced when the cutting tool is able to make a continuous cut along the workpiece. They are typically seen in materials that are ductile and have a low shear strength.

Built-up edge chips: Built-up edge chips occur when material from the workpiece adheres to the cutting edge of the tool and forms a small ridge. This ridge then breaks off from the workpiece, resulting in a chip. Built-up edge chips are typically seen in materials that are tough and have a high shear strength.

Segmented chips: Segmented chips are produced when the cutting tool cuts the material in small segments, resulting in chips that are short and stubby. They are typically seen in materials that are brittle and have a high shear strength.

Discontinuous chips: Discontinuous chips are produced when the cutting tool breaks the material into small pieces instead of cutting it continuously. They are typically seen in materials that are hard and have a low ductility.

Orthogonal Cutting Orthogonal cutting is a type of machining process in which a cutting tool is applied to a workpiece at a right angle to the direction of motion. In this process, the cutting edge of the tool removes material from the workpiece in a straight line, resulting in a flat surface.

During orthogonal cutting, the cutting tool is forced against the workpiece with a specific force called the cutting force. The cutting force is generated by the resistance of the material being cut, as well as the friction between the tool and the workpiece. The cutting force, in turn, generates a shear force that causes the material to deform and ultimately be removed from the workpiece.

Orthogonal cutting can be used to produce flat surfaces, as well as to create specific shapes and contours. The process is often used in metalworking, but can also be used with other materials such as plastics and ceramics. The parameters of the cutting process, such as the cutting speed, feed rate, and depth of cut, can be adjusted to optimize the performance and efficiency of the process.

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Oblique cutting:

Oblique cutting is a type of cutting operation in machining where the cutting tool is inclined at an angle to the workpiece surface. In contrast to orthogonal cutting, where the cutting edge is perpendicular to the workpiece surface, in oblique cutting, the cutting edge is tilted relative to the surface.

When the cutting tool is inclined at an angle, the chip formation mechanism changes, and the cutting forces, surface finish, and tool life are affected. The cutting force in oblique cutting is resolved into two components: one parallel to the cutting edge, and the other perpendicular to it. The component of the cutting force parallel to the cutting edge produces the chip formation, while the component perpendicular to the cutting edge causes the tool to bend.

Oblique cutting is used in several machining processes, such as turning, milling, and drilling. In turning, oblique cutting is used to produce tapered parts, while in milling, it is used to produce complex shapes. Oblique cutting is also used in drilling to produce conical holes.

Desirable properties of cutting tool materials

The properties that are desirable in cutting tool materials depend on the specific application and the materials being cut. However, some general properties that are typically considered desirable in cutting tool materials include:

Hardness: The cutting tool material must be hard enough to resist wear and deformation during cutting.

Toughness: The material must be tough enough to withstand impacts and shocks during cutting.

Wear resistance: The material should resist wear caused by abrasion, adhesion, and diffusion.

Heat resistance: The material should be able to withstand the heat generated during cutting without losing its properties.

Chemical stability: The material should be resistant to chemical reactions with the material being cut.

Thermal conductivity: The material should have good thermal conductivity to dissipate the heat generated during cutting.

Edge retention: The material should be able to maintain a sharp edge for a longer time.

Examples of cutting tool materials that exhibit some of these desirable properties include:

High-speed steel (HSS): A type of steel that contains tungsten, molybdenum, or cobalt, and has high hardness, wear resistance, and toughness.

Carbide: A material made of tungsten carbide particles bonded together with a metallic binder. Carbide has high hardness and wear resistance.

Ceramic: A material that is extremely hard and wear-resistant, but also brittle. Ceramics are often used in high-speed cutting applications.

Diamond: A super-hard material that is used for cutting extremely hard and abrasive materials such as stone, glass, and ceramics.

Cutting tool nomenclature:

Cutting tool nomenclature generally includes the following terms:

Shank: The shank is the part of the tool that fits into the machine's tool holder.

Flutes: Flutes are the grooves or channels in the tool that allow chips to be removed from the workpiece.

Cutting Edge: The cutting edge is the sharp edge of the tool that actually cuts into the workpiece.

Point Angle: The point angle is the angle between the two cutting edges at the tip of the tool.

Helix Angle: The helix angle is the angle between the flute and the axis of the tool.

Relief Angle: The relief angle is the angle between the face of the tool and a line perpendicular to the workpiece.

Cutting tool signature

A cutting tool signature is a set of symbols that represents important information about a cutting tool, typically used in manufacturing processes such as machining. The information conveyed by the cutting tool signature includes the type of tool, its dimensions, and the material it is made from.

The cutting tool signature typically consists of a series of numbers and letters that are arranged in a specific order. For example, a typical cutting tool signature might include the following information:

Tool type: A letter or series of letters that indicate the type of cutting tool. For example, "T" might indicate a turning tool, while "E" might indicate an end mill.

Tool dimensions: A series of numbers that indicate the dimensions of the cutting tool, such as its diameter, length, and width.

Material: A letter or series of letters that indicate the material the tool is made from, such as "H" for high-speed steel or "C" for carbide.

Single point cutting tool

A single point cutting tool is a tool used in manufacturing processes, particularly in metalworking and woodworking. It is a type of cutting tool that has only one cutting edge, which is used to remove material from the workpiece.

Single point cutting tools come in a variety of shapes and sizes, but they all share some common features. They are typically made from high-speed steel, carbide, or ceramic materials, and they are designed to be held in a tool holder that is attached to a machine tool, such as a lathe or a milling machine.

The cutting edge of a single point cutting tool is ground to a specific geometry, which is determined by the material being machined and the type of cut being made. The cutting edge may be straight or curved, and it may have one or more angles that are designed to optimize the cutting process.

When a single point cutting tool is used, the cutting edge is brought into contact with the workpiece, and the machine tool is used to move the tool across the surface of the workpiece. As the tool moves, it removes material from the workpiece in the form of chips.

Single point cutting tools are commonly used in manufacturing processes because they are precise, efficient, and versatile. They can be used to create a wide variety of shapes and sizes, and they can be used to machine a wide variety of materials, including metals, plastics, and composites.

Tool life: Tool life in machining refers to the amount of time that a cutting tool can be used before it becomes worn or damaged to the point where it can no longer effectively machine the workpiece. The length of tool life depends on a number of factors, including the type of material being machined, the type of cutting tool being used, the cutting conditions (such as speed, feed, and depth of cut), and the quality of the tool and the machine being used.

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Factors effecting the tool life

Tool life is the amount of time that a tool can be used before it becomes too worn or damaged to function effectively. The following are some of the factors that can affect the tool life:

Material of the tool: The material from which the tool is made is an important factor in determining its life. Tools made from harder and more wear-resistant materials tend to last longer.

Cutting speed: The cutting speed refers to how fast the tool is moving relative to the workpiece. A higher cutting speed can result in a shorter tool life because it increases the amount of friction and heat generated.

Feed rate: The feed rate is the speed at which the workpiece is moved into the tool. A higher feed rate can also result in a shorter tool life because it can cause more wear and tear on the tool.

Depth of cut: The depth of cut is the distance that the tool is plunged into the workpiece. A deeper cut can result in a shorter tool life because it requires the tool to do more work and generate more heat.

Different ways of measuring cutting tool

There are several ways to measure cutting tools, including:

Overall Length: This is the length of the entire tool from end to end.

Shank Diameter: This is the diameter of the part of the tool that fits into the tool holder.

Flute Length: This is the length of the flutes or grooves on the tool that allow chips to be removed from the workpiece.

Cutting Diameter: This is the diameter of the cutting edge of the tool.

Angle of the Cutting Edge: This is the angle between the cutting edge and the workpiece.

Helix Angle: This is the angle that the flutes make with the axis of the tool.

Nose Radius: This is the radius at the end of the tool where the cutting edge meets the workpiece.

Desirable properties of cutting fluids

Cutting fluids, also known as cutting oils or coolants, are used in metalworking operations to reduce friction, dissipate heat, and improve tool life. The desirable properties of cutting fluids include:

Lubricity: Cutting fluids must have good lubricating properties to reduce friction and wear between the cutting tool and workpiece. This helps to prevent tool breakage and extend tool life.

Cooling: Cutting fluids must be able to dissipate heat generated during cutting operations. Excessive heat can cause tool wear, workpiece deformation, and dimensional inaccuracies. Good cooling properties are essential for high-speed machining operations.

Chemical stability: Cutting fluids must maintain their chemical stability over time to avoid the build-up of harmful contaminants, such as bacteria and fungi. These contaminants can cause machine corrosion, unpleasant odors, and health hazards.

Compatibility: Cutting fluids must be compatible with the materials being machined, as well as with the machine tools and other components of the machining system. Incompatibility can cause chemical reactions, machine damage, and product defects.

Low viscosity: Cutting fluids must have low viscosity to facilitate their flow through the machining system, including the cutting tool, chip removal system, and coolant supply lines. Low viscosity fluids also reduce drag and improve tool life.

Purpose of cutting fluids

Cutting fluids are used in metalworking to lubricate and cool the cutting tool and workpiece during machining operations such as cutting, drilling, milling, and grinding. The primary purpose of cutting fluids is to reduce the heat generated by the cutting process, which can cause damage to the cutting tool and workpiece, and can also result in poor surface finish and dimensional accuracy. Cutting fluids also provide lubrication, which reduces friction between the cutting tool and workpiece, and can improve tool life and workpiece surface finish. In addition, cutting fluids can help remove chips and debris from the cutting zone, which can prevent the formation of burrs and other defects in the finished part.

Examples of cutting fluid

Water-based fluids: Water-based fluids are the most common type of cutting fluids. They are economical, environmentally friendly, and offer good lubrication and cooling properties. They can be used for a wide range of materials, from ferrous and non-ferrous metals to plastics.

Soluble oil fluids: Soluble oil fluids are a combination of mineral oil and water. They offer good lubrication, cooling, and corrosion protection properties. They are mainly used for cutting ferrous metals and alloys.

Synthetic fluids: Synthetic fluids are made from synthetic compounds and offer excellent cooling and lubrication properties. They are ideal for high-speed cutting operations and are suitable for cutting a wide range of materials.

Semi-synthetic fluids: Semi-synthetic fluids are a combination of synthetic and mineral oil-based fluids. They offer good lubrication and cooling properties and are suitable for both ferrous and non-ferrous material.

UNIT-4

Basic components and their functions of centre Lathe. A center lathe, also known as an engine lathe, is a machine tool used for shaping, cutting, and drilling metal or wood workpieces. It consists of several basic components, each with a specific function:

Bed: The bed is the base of the lathe that supports all the other components. It is usually made of cast iron and provides a stable platform for the workpiece and cutting tool.

Headstock: The headstock is located on the left end of the bed and contains the main spindle. It is responsible for providing power to the cutting tool and holding the workpiece in place. The spindle can rotate at various speeds, which can be adjusted to suit the type of material being machined.

Tailstock: The tailstock is located on the right end of the bed and provides support to the other end of the workpiece. It can be moved along the bed to adjust the length of the workpiece and can be locked in place once set.

Carriage: The carriage moves along the bed and holds the cutting tool. It can be moved manually or automatically using a power feed mechanism. The carriage also holds the tool post, which holds the cutting tool in place.

Cross slide: The cross slide moves perpendicular to the bed and allows the cutting tool to move across the workpiece. It is usually mounted on the carriage and can be adjusted to control the depth of cut.

Chuck: The chuck is used to hold the workpiece securely in place on the spindle. It can be tightened or released using a chuck.

Feed rod: It is a rod that runs parallel to the lead screw and is used for moving the carriage perpendicular to the axis of the lathe, which determines the depth of cut.

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Lead screw: It is a long threaded rod that runs along the bed and connects the carriage to the tailstock. It is used for moving the carriage along the bed and cutting the workpiece to the desired length.

Classification of Lathe:

Lathes can be classified based on various criteria, including:

Based on workpiece orientation:

Horizontal lathe: The spindle is mounted horizontally, and the workpiece is placed on the bed horizontally.

Vertical lathe: The spindle is mounted vertically, and the workpiece is placed on the bed vertically.

Based on bed type:

Flat bed lathe: The bed is flat and horizontal.

Slant bed lathe: The bed is angled, usually at 30 degrees, to provide better chip removal and improved rigidity.

Based on the number of axes:

2-axis lathe: It can move the workpiece in two directions, typically x-axis and z-axis.

3-axis lathe: It can move the workpiece in three directions, typically x-axis, y-axis, and z-axis.

Based on the level of automation:

Manual lathe: The operator manually controls the cutting tool's movement and feeds.

CNC lathe: It is a computer-controlled machine that uses a program to control the tool's movement and feed.

Based on the size and capacity:

Micro lathe: It is a small-sized lathe used for delicate and precise work.

Heavy-duty lathe: It is a large-sized lathe used for heavy-duty machining.

Specification of Centre Lathe.:

A center lathe, also known as a metalworking lathe or engine lathe, is a machine tool that rotates a workpiece about its axis to perform various operations such as cutting, drilling, facing, and turning.

Here are some of the specifications of a center lathe:

Swing over bed: This is the maximum diameter of the workpiece that can be accommodated by the lathe. It is measured by the distance from the bed to the center of the spindle multiplied by two. Typical swing over bed sizes range from 9 inches to 60 inches.

Distance between centers: This is the maximum length of the workpiece that can be supported between the headstock and tailstock centers. Typical distances range from 20 inches to 120 inches.

Spindle bore: This is the diameter of the hole in the spindle through which the workpiece is fed. Typical spindle bore sizes range from 1 inch to 6 inches.

Spindle speeds: The spindle speed range is the number of rotations per minute (RPM) that the lathe can achieve. Most center lathes have a range of speeds that can be adjusted to suit the material being machined. The range typically varies from 50 to 3000 RPM.

Motor power: This is the power rating of the electric motor that drives the spindle. Motor power typically ranges from 1 HP to 20 HP.

Bed length: This is the length of the bed that supports the headstock, tailstock, and carriage. Bed lengths typically range from 30 inches to 240 inches.

Carriage travel: The carriage is the part of the lathe that moves along the bed and holds the cutting tool. Carriage travel is the maximum distance that the carriage can travel along the bed. Typical carriage travel ranges from 4 inches to 40 inches.

Cross slide travel: This is the maximum distance that the cross slide, which holds the cutting tool, can move perpendicular to the axis of the lathe. Typical cross slide travel ranges from 2 inches to 12 inches.

Tailstock travel: The tailstock is the part of the lathe that supports the other end of the workpiece. Tailstock travel is the maximum distance that the tailstock can move along the bed to accommodate different workpiece lengths. Typical tailstock travel ranges from 4 inches to 20 inches.

Taper attachment: A taper attachment is an optional accessory that allows the lathe to cut tapered workpieces. The taper attachment can be set at different angles to produce different taper angles.

Comparison among Centre Lathe, Capstan Lathe and Turret Lathe.

Feature	Centre Lathe	Turret Lathe	Capstan Lathe
Headstock	Cons pulley or all gear driven. Can Provide lesser speed than the other two	Heavy construction, wider range of spindle speeds from more powerful motor than centre lathes	Same as Turret Lathe
Toolpost	Usually a single Toolpost (with a single tool). Sometimes, a square turret type Toolpost is provided which can hold 4 tools.	Its cross slide carries a square turret type Toolpost carrying 4 tools. Additionally, it has a Toolpost at its rear to hold 1 or 2 tools.	Same as Turret Lathe
Tailstock	Usually used for supporting the WP. Sometimes, some cutting tools (drills, reamers etc) installed to perform operations at the end face of WP.	Carries a hexagonal *turret head* instead of tailstock, which may hold a tool or tool head on each of its face. They are automatically indexed in sequence to bring each tool in proper position for performing operations sequentially.	Carries a square, round or hexagonal turret called *Capstan head* in place of tailstock. This provides the same facility of mounting and indexing 4-6 tools as in case of turret lathe.
Leadscrew	Always provided on centre lathe to enable thread cutting by a single point tool.	Thread cutting generally performed by taps and die heads fitted to the turret head.	Same as Turret Lathe
Method of mounting the *Turret* or *Capstan* Head	Not applicable	*Turret head* directly mounted on a saddle and for feeding the tool on the WP the entire saddle has to be moved.	*Capstan head* is mounted on a *ram* or *slide*, which travels on the fixed saddle. The tools are fed on the WP by moving the *slide*.
No. of tools mounted	Usually 1 for single tool post. If a square turret type Toolpost is provided then 4 tools can be mounted.	4 tools on the front tool post, 1 or 2 on the rear tool post and 4-6 tools on the square or hexagonal *turret* head. No. of tools can be increased by using multiple tool holders on the *turret*.	Same as Turret Lathe
Skill required for operator	Very high	Very nominal after setting of tools	Same as Turret Lathe
Tool setting time	High. Tool has to be changed every time a new operation is to be	Initial tool setting time required. During operation all the tool are	Same as Turret Lathe

Lathe accessories:

Chucks: A chuck is a clamping device that is used to hold the workpiece in place while it is being turned on the lathe. There are different types of chucks, including 3-jaw and 4-jaw chucks, which can hold round, square or hexagonal workpieces.

Mandrel: A mandrel is a cylindrical device that is used to support the inside of hollow workpieces while they are being turned on the lathe. It is often used in conjunction with a lathe chuck to hold the workpiece in place.

Rests: Rests are devices that are used to support the workpiece as it is being turned. There are different types of rests, including the tool rest, which supports the cutting tool, and the steady rest, which supports long, thin workpieces to prevent them from vibrating.

Face plates: A face plate is a flat metal plate that is attached to the lathe spindle and is used to hold large, flat workpieces that cannot be held in a chuck. The workpiece is bolted onto the faceplate and then turned using the lathe.

Centers: Centers are devices that are used to support the workpiece at the ends while it is being turned. There are different types of centers, including the live center, which rotates with the workpiece, and the dead center, which remains stationary.

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Angle plates: Angle plates are metal plates with a flat surface and a 90-degree angle that are used to hold the workpiece at a specific angle to the lathe axis. They are often used in conjunction with a lathe chuck or faceplate.

Taper turning on Lathe:

Tapper turning on a lathe refers to the process of creating a tapered shape on a workpiece using a lathe machine. This is done by adjusting the angle of the lathe's cutting tool and the position of the tailstock, which holds the other end of the workpiece.

1. To turn a taper on a lathe, you will need to follow these steps:
2. Mount the workpiece securely between the lathe's chuck and the tailstock.
3. Adjust the angle of the lathe's cutting tool to match the desired taper angle.
4. Move the cutting tool towards the workpiece and begin to make light cuts.
5. Gradually move the cutting tool closer to the tailstock as you continue to make cuts. This will increase the diameter of the workpiece at the tailstock end and create a tapered shape.
6. Continue to make cuts until the desired taper shape is achieved.

Different ways of representing taper on a job:

Taper refers to the gradual decrease in diameter or width of a cylindrical or conical shape. When working on a job that involves taper, there are several ways to represent it:

Taper angle: This is the angle at which the taper deviates from a straight line. It is usually measured in degrees and can be represented as a symbol on a drawing, such as a triangle.

Taper per inch (TPI): This is the ratio of the change in diameter or width to the length of the tapered section. It is typically represented as a fraction, such as 1:12, where the first number represents the length of the taper and the second number represents the change in diameter or width.

Taper ratio: This is similar to TPI but is expressed as a decimal instead of a fraction. It represents the ratio of the diameter or width at one end of the tapered section to the diameter or width at the other end.

Taper length: This is the length of the tapered section and is typically measured in inches or millimeters. It can be represented as a dimension on a drawing.

Taper per foot (TPF): This is similar to TPI but is expressed as the change in diameter or width per foot of length. It is typically represented as a fraction, such as 1/8 inch per foot.

Tapered radius: This is the radius of the tapered section and can be represented as a dimension on a drawing.

These are some of the most common ways to represent taper on a job.

Different methods of taper turning

Taper turning is the process of creating a gradual decrease in diameter along the length of a cylindrical workpiece. There are several methods for taper turning, including:

Offset Tailstock Method: In this method, the tailstock is offset by a calculated amount to create the desired taper angle. The workpiece is then turned using the standard method, and the offset tailstock compensates for the taper angle.

Compound Rest Method: In this method, the compound rest is set at an angle to the lathe bed, which allows the cutting tool to move at an angle to the axis of the workpiece. The angle of the compound rest is adjusted to create the desired taper angle.

Taper Turning Attachment: A taper turning attachment is a specialized tool that is attached to the lathe and allows the cutting tool to move at an angle to the axis of the workpiece. The angle of the attachment is adjusted to create the desired taper angle.

Form Tool Method: In this method, a form tool with the desired taper angle is used to cut the workpiece. The form tool is mounted on a tool post and fed into the workpiece at the appropriate angle.

Graduated Collar Method: In this method, a graduated collar is used to set the taper angle. The collar is mounted on the tailstock and is adjusted to the desired angle. The workpiece is then turned using the standard method, and the collar compensates for the taper angle.

Thread cutting on Lathe:

Procedure:

Thread cutting is a common operation performed on a lathe to produce threads on a workpiece. The following is a general procedure for cutting threads on a lathe:

Choose the correct cutting tool: The cutting tool should be appropriate for the material being threaded and the thread size to be cut.

Set the tool height: The cutting tool should be set at the correct height to ensure the correct thread pitch is achieved.

Set the thread depth: The thread depth should be set based on the desired thread pitch and depth of the thread.

Set the lathe speed: The lathe speed should be set to an appropriate speed for the material being threaded and the thread size being cut.

Position the tool: Position the cutting tool at the starting point of the thread.

Engage the lead screw: Engage the lead screw and advance the carriage towards the workpiece.

Start cutting: Begin cutting the thread by turning the lathe's spindle.

Move the tool: Move the cutting tool along the workpiece to cut the desired length of the thread.

Disengage the lead screw: Disengage the lead screw and return the carriage to its starting position.

Check the thread: Check the thread pitch and depth with a thread gauge.

Repeat as necessary: Repeat the process until the desired number of threads have been cut.

Change gears calculation for thread cutting operation :

The gear ratio required for thread cutting depends on the number of threads per inch (TPI) of the thread you wish to cut, as well as the pitch of the lead screw on your lathe. The formula for calculating the gear ratio required for thread cutting is:

Gear Ratio = (Number of Threads Per Inch) x (Pitch of Lead Screw)

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Lathe operations:

Centering: This is the process of aligning the workpiece with the lathe's axis of rotation. This is usually done using a center drill or live center, and ensures that the workpiece spins true and produces accurate cuts.

Facing: This operation involves cutting a flat surface on the end of the workpiece, typically to prepare it for other operations such as turning or drilling. A facing tool is used for this operation.

Turning: This is the process of removing material from the workpiece to create a cylindrical shape. A cutting tool is moved along the length of the workpiece to create the desired shape and size.

Parting Off: This is the process of cutting the workpiece off from the rest of the stock. A parting tool is used for this operation.

Undercutting: This is the process of creating a groove or recess in the workpiece, usually for the purpose of accommodating a snap ring or other retaining device. An undercutting tool is used for this operation.

Grooving: This is the process of cutting a groove into the workpiece, typically for the purpose of accommodating a keyway or other feature. A grooving tool is used for this operation.

Knurling: This is the process of creating a pattern of small ridges on the surface of the workpiece, typically to improve grip or appearance. A knurling tool is used for this operation.

Drilling: This is the process of creating a hole in the workpiece. A drill bit is used for this operation.

Boring: This is the process of enlarging an existing hole in the workpiece to a specific size and depth. A boring bar is used for this operation.

Cutting parameters:

Cutting parameters are the set of conditions or settings used during machining operations to determine the optimal speed, feed rate, depth of cut, and tool selection for a particular material and cutting tool. These parameters directly affect the quality, efficiency, and cost of the machining process.

The cutting speed is the surface speed of the workpiece in meters per minute or feet per minute. The feed rate is the distance the cutting tool moves in the workpiece per revolution, typically measured in millimeters per tooth or inches per tooth. The depth of cut is the amount of material removed from the workpiece in a single pass, typically measured in millimeters or inches. Tool selection involves choosing the most appropriate cutting tool based on the material being machined and the desired finish.

The cutting parameters must be optimized for each material and tool combination to achieve the best results in terms of cutting time, tool life, surface finish, and dimensional accuracy. In general, higher cutting speeds and feed rates result in faster cutting times, but may also lead to shorter tool life and poorer surface finish. Smaller depth of cut is generally associated with higher surface finish, but slower machining time.

Cutting parameters can be adjusted manually by the machinist, or automatically by a computer numerical control (CNC) system. The optimal cutting parameters will depend on factors such as the material being machined, the cutting tool being used, the machine tool being used, and the desired outcome of the machining process.

UNIT-5

Types of pattern:

Single-piece pattern: A pattern that is made in one piece and used to create a single mold.

Split pattern: A pattern that is made in two or more pieces, which are joined together to create the mold.

Match plate pattern: A type of split pattern in which the two halves are mounted on opposite sides of a plate, called a match plate. This allows for faster and easier production of molds.

Sweep pattern: A pattern that is used to create a curved or irregular shape in a casting.

Loose-piece pattern: A pattern in which some pieces can be removed from the mold before casting, allowing for the creation of undercuts and other complex shapes.

Cope and drag pattern: A pattern in which the mold is made in two halves, with the top half (cope) and the bottom half (drag) being made separately.

What is pattern?

In casting, a pattern refers to a replica of the final product or part that is to be cast in metal, typically made from wood, metal, plastic, or other materials. The pattern is used to create a mold into which molten metal is poured to create the final casting.

Pattern allowance:

Pattern allowance refers to the additional material or space added to a part or component in a manufacturing process to compensate for dimensional variations caused by the manufacturing process itself, such as shrinkage, warping, or deformation.

Types of pattern allowances:

There are several types of pattern allowances used in manufacturing processes, including:

Shrinkage Allowance: This is the amount of additional material added to a pattern to compensate for the expected shrinkage of the material during the cooling process. Shrinkage allowance is commonly used in casting and molding processes.

Machining Allowance: This is the additional material added to a pattern to compensate for the material that will be removed during the machining process. Machining allowance is commonly used in milling, turning, and drilling operations.

Draft Allowance: This is the additional taper added to the vertical walls of a pattern to allow for the easy removal of the pattern from the mold. Draft allowance is commonly used in injection molding and casting processes.

Fillet Allowance: This is the additional curved surface added to the internal corners of a pattern to prevent sharp corners in the finished product. Fillet allowance is commonly used in casting processes.

Warpage Allowance: This is the additional material added to a pattern to compensate for the expected warpage of the material during the cooling process. Warpage allowance is commonly used in injection molding and other plastic forming processes.

Pattern materials:

Patterns material refers to a type of fabric or textile that features a repeating design or motif. These designs can vary widely, from simple geometric shapes to complex floral arrangements or intricate abstract designs. The patterns can be woven directly into the fabric or printed onto it using a variety of techniques such as screen printing, digital printing, or block printing. Patterns materials are often used in

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clothing, home decor, and other textile products to add visual interest and create a unique look. They can be made from a variety of materials, including cotton, silk, wool, and synthetic fibers, and come in a wide range of colors, textures, and styles.

Types of pattern materials:

There are various types of patterns materials used in various fields, including:

Wood: Wood is one of the most commonly used materials for pattern making. It is easy to work with and can be carved or machined to create intricate designs.

Plastic: Plastic patterns are lightweight and can be easily produced using injection molding or 3D printing techniques. They are durable and can be used to create complex shapes.

Metal: Metal patterns are typically made from aluminum or steel and are used for high-volume production. They are durable and can withstand the rigors of the manufacturing process.

Wax: Wax patterns are commonly used in the jewelry industry for lost-wax casting. They are melted away during the casting process, leaving behind a precise metal replica of the original design.

Foam: Foam patterns are lightweight and easy to carve, making them a popular choice for creating prototypes or one-off designs.

Silicone: Silicone rubber is used to create flexible, durable patterns that can be used for casting a wide range of materials, including concrete, plaster, and resin.

Moulding materials:

Moulding materials are materials that are used to create a specific shape or form by pouring or pressing the material into a mould. The moulding material can be any substance that can be poured, pressed or otherwise shaped into a specific form or shape, and then allowed to harden or cure.

Types of moulding sand:

There are several types of moulding sand used in the foundry industry. Here are some common types:

Green sand: Green sand is the most common type of moulding sand used in foundries. It is made from a mixture of silica sand, bentonite clay, water, and additives.

Dry sand: Dry sand is similar to green sand, but it is dried and baked to remove moisture. This makes it more stable and better suited for certain types of casting.

Loam sand: Loam sand is a mixture of sand, clay, and organic materials. It is used for large castings and where a smooth surface finish is required.

Facing sand: Facing sand is a fine-grained sand used for the surface of the mould. It is used to improve the surface finish of the casting.

Backing sand: Backing sand is used to fill the flask and support the mould during pouring. It is made from coarser sand than facing sand.

Core sand: Core sand is used to create cores, which are inserts placed inside the mould to create cavities or complex shapes in the casting. It is made from a mixture of sand, clay, and a binder.

Oil sand: Oil sand is a type of moulding sand that contains oil as a binder. It is used for casting metals that require a high degree of accuracy and a smooth surface finish.

Properties of sand:

Refractoriness: Refractoriness refers to the ability of sand to withstand high temperatures without melting or degrading. The sand used in moulding should have a high refractoriness, which is essential for producing high-quality castings.

Permeability: Permeability refers to the ability of sand to allow gases to escape from the mould. The sand should have adequate permeability to allow the gases generated during casting to escape freely without creating voids or defects in the casting.

Cohesiveness: Cohesiveness refers to the ability of sand to stick together when compacted. The sand used in moulding should have good cohesiveness to form a solid mould that can withstand the pressure of molten metal without collapsing.

Plasticity: Plasticity refers to the ability of sand to be shaped into a specific form without cracking or breaking. The sand used in moulding should have adequate plasticity to be easily shaped into the desired mould.

Strength: The sand used in moulding should have adequate strength to hold its shape under the weight of molten metal. This property is important for producing accurate and defect-free castings.

Moulding methods:

Molding is a manufacturing process where a material, often in a liquid or semi-liquid state, is shaped using a mold. There are many different types of molding methods, each with their own advantages and disadvantages. Here are some of the most common molding methods:

Injection molding: This is the most common method for producing plastic parts. Molten plastic is injected into a mold under high pressure, then allowed to cool and harden.

Blow molding: Used to create hollow plastic parts, such as bottles or containers. A tube of molten plastic is inflated inside a mold, then allowed to cool and solidify.

Compression molding: This method is often used for producing large parts or parts with complex shapes. A material is placed in a heated mold, then compressed under high pressure until it takes the shape of the mold.

Rotational molding: Used for producing large, hollow plastic parts, such as tanks or playground equipment. A mold is rotated while a material is heated and distributed evenly, then allowed to cool and solidify.

Extrusion molding: This method is used for producing long, continuous parts with a consistent cross-section, such as pipes or tubing. A material is forced through a die, which shapes it into the desired shape.

Thermoforming: Used for producing plastic parts with a variety of shapes and sizes. A sheet of plastic is heated until it becomes pliable, then formed into a mold using vacuum or pressure.

Core: A core is a body which is set into the prepared mould before pouring the molten metal to form holes,recesses,under cuts,internal cavities etc.

Core point : A core point can refer to a central or essential aspect of something. It is a fundamental idea or concept that is crucial to understanding the subject matter or achieving a goal. In the context of a discussion or argument, a core point can be a key argument or position that serves as the foundation for the overall perspective or stance. A core point can also refer to a crucial or critical element of a plan or strategy that is necessary for its success. In essence, a core point is a critical component that cannot be overlooked or disregarded.

Elements of Gating system:

The gating system in casting refers to the network of channels and openings that allow molten metal to flow from the pouring cup to the mold cavity. The gating system has several elements that are crucial for ensuring proper filling of the mold and producing high-quality castings. These elements include:

Pouring cup: This is the container used to pour the molten metal into the gating system. It is typically made of ceramic, refractory material, or steel.

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Sprue: The sprue is a vertical channel that connects the pouring cup to the runners. It allows the molten metal to flow from the pouring cup to the runners.

Runners: Runners are horizontal channels that connect the sprue to the gates. They distribute the molten metal evenly to each gate.

Gates: The gates are the openings in the mold cavity where the molten metal enters. They control the flow of metal into the cavity and help to prevent turbulence.

Riser: A riser is a reservoir that holds extra molten metal to feed the casting during solidification. It helps to prevent shrinkage defects in the casting.

Filter: A filter is used to remove any impurities or contaminants in the molten metal before it enters the mold cavity. This helps to ensure the quality of the casting.

Bench Moulding:

Bench molding is a process used in foundry work to create sand molds for casting metal objects. There are several bench molding methods, including:

Sweep or straight molding: This is the most basic method of bench molding, where a flat board called a sweep is used to level the sand and create a flat surface. The pattern is then placed on the sand and more sand is added and compacted around it.

Rollover molding: This method involves using a two-piece flask (a container for the sand) that is rolled over to create an impression in the sand. The pattern is placed on one side of the flask, and sand is added to the other side. The flask is then rolled over onto the pattern, creating the mold.

Drag and cope molding: This method involves using a two-piece flask where one half is called the drag and the other is the cope. The pattern is placed on the drag, and sand is added and compacted around it. The cope is then placed on top of the drag, and sand is added and compacted around it.

Match plate molding: This method uses a match plate, which is a metal plate that has the pattern for both the top and bottom of the mold. Sand is packed onto the plate, and then the plate is flipped over and more sand is added to the other side.

Shell molding: This method involves using a mixture of sand and resin to create a thin shell mold. The pattern is placed on the sand, and then the mixture is poured over it. The mold is then removed from the pattern and baked in an oven to cure the resin.

Floor moulding methods

Floor moulding, also known as baseboard or skirting, is a finishing trim used to cover the joint between the wall and the floor. Here are a few common methods for installing floor moulding:

Nail gun: One of the most popular methods for installing floor moulding is using a nail gun. The moulding is held in place against the wall, and the nail gun is used to attach it to the wall. This method is fast and efficient, but requires a bit of skill to ensure that the nails are driven in at the correct angle.

Finish nails and hammer: Another method for installing floor moulding is using finish nails and a hammer. This method is slower and more labor-intensive than using a nail gun, but it is also more precise, and there is less risk of damaging the moulding.

Adhesive: For a clean, seamless look, some people choose to use adhesive to attach the floor moulding to the wall. This method is popular for modern and minimalist design styles. However, it can be tricky to get the moulding to stick in the right place, and it may not be as secure as using nails.

Combination of methods: Some installers use a combination of the above methods to ensure a secure and seamless installation. For example, they may use adhesive to hold the moulding in place while the nails or screws are driven in.

Furnace: Furnaces in casting are heating devices used to melt and maintain the temperature of metals or alloys in their molten state for casting. The furnaces are designed to reach and maintain high temperatures necessary for melting different metals and alloys, and they can be electrically or fuel-powered.

construction and working of cupola furnace:

A cupola furnace is a type of melting furnace used for melting cast iron. It is a tall, cylindrical furnace made of steel plates lined with refractory materials such as fireclay, silica, or alumina.

The construction of a cupola furnace includes the following components:

Stack: The stack is a tall, cylindrical structure that provides the draft necessary for the combustion of the fuel.

Tuyeres: Tuyeres are openings in the bottom of the furnace through which air is blown into the furnace to support the combustion process.

Charging door: The charging door is a large opening on the side of the furnace through which the raw materials (i.e. iron scrap, coke, and limestone) are charged into the furnace.

Slag hole: The slag hole is a small opening at the base of the furnace used for draining off the waste material, called slag, that forms during the melting process.

The working of a cupola furnace can be divided into the following stages:

Charging: The raw materials, including scrap iron, coke, and limestone, are loaded into the furnace through the charging door.

Ignition: The coke is ignited and burns to create heat, which melts the iron scrap.

Melting: The molten iron is collected at the bottom of the furnace, where it is tapped off periodically into ladles for casting.

Slag removal: As the iron is melted, the impurities in the raw materials combine with the limestone to form slag. The slag is drained off periodically through the slag hole.

Shutting down: Once the desired amount of iron has been melted, the furnace is shut down by stopping the flow of air and allowing the coke to burn out.

construction and working of electric furnace:

Electric arc furnaces are a type of industrial furnace used for melting metals and alloys. They use an electric arc to heat the material and create the high temperatures needed for melting.

Construction:

Electric arc furnaces consist of a refractory-lined vessel that is usually shaped like a cylinder or a box. The vessel is mounted on a base, which contains a mechanism for tilting the furnace to pour out the molten metal. The vessel is usually made of steel, and the refractory lining is made of materials such as magnesite, dolomite, or alumina.

The furnace is equipped with three graphite electrodes, which are inserted through the roof of the furnace. The electrodes are connected to an electrical power source through cables, and they can be moved up or down to control the intensity of the electric arc.

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Working:
The electric arc furnace works by creating an electric arc between the electrodes and the material being melted. The arc produces high temperatures, typically around 1600-1800°C (2900-3300°F), which melt the material.
To start the melting process, scrap metal or other materials are loaded into the furnace. Once the material is in place, the electrodes are lowered until they are just above the material. An electric current is then passed through the electrodes, creating an arc that heats the material to the melting point.
As the material melts, impurities such as slag and gases are produced. The slag floats on top of the molten metal and is removed through a slag door in the side of the furnace. Gases are removed through a vent in the roof of the furnace.
Once the melting process is complete, the furnace is tilted to pour out the molten metal into a ladle or other container for further processing.

Die casting
Die casting is a manufacturing process in which molten metal is injected under high pressure into a steel mold, or die, to produce complex and detailed parts. The die casting process is commonly used to produce a wide range of parts for various industries such as automotive, aerospace, electronics, and consumer goods.
The die casting process involves several steps:
Mold Preparation: The steel mold, or die, is prepared by cleaning and coating with a release agent to prevent the molten metal from sticking to the mold.
Melting and Injection: The metal, usually an alloy of aluminum, zinc, or magnesium, is melted in a furnace and then injected into the mold under high pressure. The pressure helps to fill the mold cavity completely and evenly.
Cooling: Once the molten metal has been injected into the mold, it is left to cool and solidify. Cooling time depends on the size and complexity of the part.
Ejection: After the part has cooled and solidified, the mold is opened, and the part is ejected from the mold.
Trimming and Finishing: The part may have excess material or flash that needs to be removed. Trimming and finishing operations such as sandblasting, polishing, and painting are done to achieve the desired final product.

centrifugal casting:
Centrifugal casting is a casting process used to manufacture a wide range of cylindrical parts such as pipes, tubes, and rings. The process involves pouring molten metal into a rotating mold or die, allowing the centrifugal force to distribute the metal evenly along the mold cavity, resulting in a denser, higher-quality casting.

The centrifugal force generated by the rotation of the mold causes the molten metal to be forced outward, resulting in the formation of a cylindrical shape with a bore hole at the center. The centrifugal force also helps to remove impurities and gas pockets from the casting, resulting in a more uniform and defect-free product.

Centrifugal casting can be performed in either a horizontal or vertical orientation. In the horizontal method, the mold is rotated around its horizontal axis, while in the vertical method, the mold rotates around its vertical axis. The vertical method is often preferred for larger parts, as it allows for greater control over the casting process and produces parts with a more uniform wall thickness.

Some common applications of centrifugal casting include the production of pipes, cylinders, and rings for use in the automotive, aerospace, and marine industries.

Investment casting:
Investment casting, also known as lost wax casting, is a manufacturing process used to create complex metal parts with high accuracy and surface quality. Here are the basic steps involved in investment casting:
Pattern Creation: A pattern, which is an exact replica of the final part, is created using wax or a similar material. This pattern is typically made by injecting wax into a mold.
Assembly: The wax patterns are then attached to a central wax sprue or gating system, which will allow molten metal to flow into the pattern during the casting process.
Shell Building: The assembly is then dipped into a ceramic slurry, which will form a shell around the wax pattern. The shell is typically built up in multiple layers to create the desired thickness.
Dewaxing: Once the shell has fully dried and hardened, it is placed in a furnace where the wax is melted out, leaving behind a cavity in the shape of the pattern.
Pouring: Molten metal is then poured into the cavity, filling the space left by the wax pattern.
Cooling: The metal is allowed to cool and solidify inside the shell.
Shell Removal: The ceramic shell is then broken away, revealing the final metal part.
Finishing: The final part is typically machined or polished to achieve the desired surface finish and dimensional accuracy.

Shell moulding:
Shell molding is a type of foundry process used to produce high-precision castings with a smooth surface finish. The process involves creating a shell mold, which is a thin-walled mold made from a mixture of sand and thermosetting resin. The shell mold is then heated to a specific temperature and then filled with molten metal.
The shell molding process typically involves the following steps:
Pattern creation: A pattern is created out of wood, plastic, or metal, which is used to form the shape of the final casting.
Shell mold creation: The pattern is placed into a metal tool called a shell box, and a mixture of sand and resin is poured around it. The resin is then cured using heat, and the shell mold is removed from the box.
Core creation: If necessary, a core made of sand or metal is inserted into the shell mold to create internal features of the casting.
Pouring: The shell mold is preheated to a specific temperature, and then molten metal is poured into it.
Cooling and release: The molten metal cools and solidifies inside the shell mold. Once the metal has solidified, the shell mold is broken away from the casting, leaving a smooth surface finish.

Casting defects:
1.**Gas defects:**
(a) Open blows:Gas defects which are formed at the outside surface.
Blow holes:Gas defects which are formed at the inside of casting.
(b) Scar: A shallow blow which is formed on the flat surface of a casting.
(c) Blister: when scar is covered by a thin layer of metal.
2.**Moulding material and methods:**
(a) Drop and Dirt : Due to improper ramning,the moulding sand will be dropped from cope to the drag box and will form a projection on the surface of the casting,known as

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Drop and a cavity at the bottom surface of the casting known as drift.
(b)Scab : Due to improper ramming the liquid metal may enter into loose sand layers in the cope box.
3.**Pouring metal defects** :
(a) Misrun: due to lack of fluidity and pouring temp. of liquid metal ,if the metal gets solid before reaching the extreme and then the defects is known as misrun.
(b) Cold shut: Two streams of liquid metals coming from two different directions will not fuse,then that defects is known as cold shut.
4. **Hot Cracks :** These defects are caused by tensile stresses that develop in the metal as it solidifies and contracts, and they can lead to a loss of strength and structural integrity in the final product.
Hot cracks typically occur in areas of the casting where the cooling rate is slow, such as thick sections or corners. As the metal cools and solidifies, it contracts and pulls away from the surrounding material, creating a tensile stress that can cause cracking.
5. **Mould Shift :** Mould shift is a casting defect that occurs when the mold used to create a casting is not properly aligned, resulting in an offset or shift between the two halves of the mold. This misalignment can cause the molten metal to flow into one side of the mold more than the other, resulting in a casting that is not uniform in shape.

Advantages of casting:
1.Complex shape of objects can be produced.
2.Less expensive and simple process.
3.It can be used to make small to very large size part.
4.Any material can be cast ferrous & non-ferrous.
5.Certain metals & alloys are produced only by castings.

Disadvantages of castings:
1.A high tooling cost and a long set-up time.
2.The permanent mould casting process is ,in general limited to smaller castings.
3.Because of the high tooling cost involved a high production volume is needed in order to make this process and economically viable manufacturing option.
4.The higher poring temperature of the molten metal the shorter the life of the mould.

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Definition of welding:
Welding is a process in which localised permanent joints can be produced with or without application of heat,with or without application of pressure or pressure alone and with or without application of filler material for joining of similar or dissimilar materials.

Arc welding:
Definition: Electric arc welding is "a welding process wherein coalescence is produced by I heating with an arc or arcs, with or without the application of pressure and with or without the use of filler metals.
Principle: Definition of electrical arc: An electric arc is a continuous stream of electrons flowing through some sort of medium between two conductors of an electric circuit and accompanied by intense heat generation and radiation. An electric arc for welding is obtained in the following ways:
1. Between a consumable electrode (which also supplies filler metal) and the work piece.
2. Between a non consumable electrode (carbon, graphite or tungsten etc.) and the work piece.
3. Between two non-consumable electrodes.

DCSP: In D.C. Straight Polarity, work is positive (anode) and electrode is cathode. Therefore, the work material is heated much faster than the electrode. This is of advantage in welding massive pieces because it puts the heat where needed. This system is preferred on all metals except aluminium, magnesium, copper and beryllium. Its advantages are , deeper penetration, the narrowest heat-affected zone, the fastest travel speed, less electrode consumption and least distortion because of the faster heat input into the work. Also, we can use bare and medium coated electrodes as they require less amount of heat for melting.

DCRP: In D.C. reverse polarity electrode is positive and the work is negative terminal. The electrons flow from the work piece which tends to clean it. But the heat flowis also towards the electrode making its consumption faster and so limiting the current that can be put through an electrode. This system is preferred in welding thin sections such as automobile bodies and non-ferrous metals to prevent the melting or burning of holes in the metal, since only about 38% of the arc energy is liberated at the work surface in this case. Because this system has an inherent ability to scour the oxide film from the surface of the work, it is very useful for welding aluminium and magnesium whose oxides are very hard and brittle. A wide bead (due to shallow penetration) and cleaner work surface are the characteristics of this system.

SHIELDED METAL ARC WELDING:
Process: The single metal arc welding is also known as manual metal arc welding is a manual metal arc welding process that uses a consumable and protected electrode.as the electrode melts,a cover that protects the electrode melts, a cover that protects the electrode melts and protects the weld area from oxygen and other atmospheric gases.
Application: SMAW can be used for a variety of metal types and various thickness. It is often used for heavy duty work involving industrial iron and steel like carbon steel and cast iron , as well as work involving low and high alloy steels and nickel alloys. SMAW is used in a variety of industries , including , Construction.

SUBMERGED ARC WELDING:
Process: SAW is a common arc welding process that involves the formation of an arc between a continuously fed electrode and the workpiece. A blanket of powdered flux generates a protective gas shield and a slag (and may also be used to add alloying elements to the weld pool) which protects the weld zone.
Application: SAW process is used for welding of low alloy steels. It is also used for stainless steel and nickel based alloys. SAW can also be used for surfacing applications such as wear-facing ,corrosion overlay of steels etc.

Tungsten Inert Gas welding:
Process: TIGW is uses the heat generated by an electric arc struck between a non consumable tungsten electrode and the work piece to fuse metal in the joint area and produce a molten weld pool.
Equipments: TIGW equipment comprises a power source , grounding cable , welding torch and shielding gas tank or gas network interface.
Application: (1) pipeline and pipe welding.
(2) Bike frames.
(3) Door handles.
(4) Lawn mowers.
(5) Fenders.

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Metal Inert Gas :
Process: Metal inert gas welding uses the heat generated by an electric struck between a consumable electrode and the workpiece.
Application: (1) Automotive.
(2) Construction.
(3) High-production manufacturing.
(4)Aerospace.
(5) Rail-road.

THERMIT WELDING :
Process: Thermit welding is a chemical welding process where heat is generated from an exothermic chemical reaction is used for the fusion. The chemical reaction aluminothermic process occurs between aluminium powder and metal oxide. This reaction generates a molten metal that acts as filler metal joining the workpieces on solidification.
Application: Thermit welding is used for repair of steel casing and forgins for joining railroad rails,steel pipes for joining large cast and forged parts.
Thermit mixture:
$3Fe_3O_4 + 8\ Al$ ---- $9Fe + 4Al_2O_3$

$Fe_3O_4 : Al = 3:1$

GAS WELDING:
PROCESS: Oxyacetylene welding , commonly reffered to as gas welding is a process which relies on conmbustion of oxygen and acetylene when mixed together in correct proportions with in a hand-held torch or blow pipe , a relatively hot flame is produced with a temperature of about 3200**°C.**
Primary reaction: C2h2+o2 ---- 2Co+ h2+ $\Delta H1$
Secondary reaction:

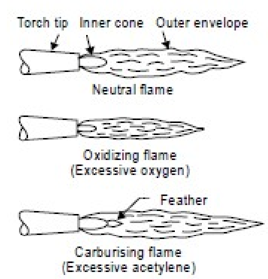
2Co+ h2+3/2 o2 ---- 2Co2 + H2o + $\Delta H2$

C_2H_2 + 5/2 o2 ---- 2Co2 + H2o + ΔH

Types of Flames:
1. Neutral Flame: (Oxygen : Acetylene = 1.0) When oxygen and acetylene are supplied in nearly equal volumes, this is produced having a max. temperature of 3200oC. This is desired in most welding operations. It has sharp brilliant Inner cone and outer cone faintly luminous with bluish colour (1250oC). Used for most welding applications for many metals like Mild steel, Stainless steel, Cast Iron, Copper, Aluminium etc.

2. Carburizing Flame: (Oxygen : Acetylene = 0.85 to 0.95) There is excess of acetylene. This has 3 zones, sharp inner cone (2900oC), intermediate whitish cone, bluish outer cone. The length of the intermediate cone is an indication of the proportion of excess acetylene. If little excess of acetylene is used it is called reducing condition and is used for welding Low carbon steel, Ni, non-ferrous Alloys, low alloy steel etc. If more excess of acetylene is used it is called carburizing condition and is used for low carbon steels for carburizing heat treatment purpose.

3. Oxidizing Flame: (Oxygen : Acetylene = 1.15 to 1.5) There is excess oxygen. It has inner cone (3300oC) with purplish tinge and outer cone. This is used for non-ferrous alloys. Such as Cu-base and Zn-base alloys like Brass (Cu- Zn) and bronze (CuSn). The oxidizing atmosphere, in these cases, creates a base metal oxide that protects the base metal. For example, in welding brass, the zinc has a tendency to separate and fume away. The formation of a covering copper oxide prevents the zinc from dissipating.



RESISTANCE WELDING :
Spot welding : Spot welding is a type of electric resistance weld that uses resistance to weld two or more metal sheets together using pressure and heat to the weld part. The spot welding process uses two copper alloy electrodes to focus the welding current into a small area and hold the sheets together.
Application: Spot welding has applications in a number of industries, including automotive, aero space,rail, white goods metal furniture., electronics, medical building and construction.

Principle : Resistance spot weldimg works on the principle of Joule law of heating, where the heat generated is directly proportional to the square of the welding current.

*leak proof joints can not be produced.

Seam welding :
Process : Resistance seam welding is a variant of the basic resistantce spot welding process. In spot welding weld nugget is produced by passing on electric current through the sheet materials to be welded while they are held together the sheet materials to be welded while they are held together under a mechanical force between shaped copper electrodes in a lap configuration.
*leak proof joints can be produced
Application :
1.It can be used to make gas or fluid tight joints in a variety of sheet metal fabrications.
2.Steel fuel tanks for motor vechiles.

Projection Welding :
Process: In resistance projection welding , small prijections are formed on one or both pieces of the base metal to obtain contact at a point which localize the current flow and concentrate the heat. Under pressure, the heated and softened projections collapse and a weld is formed.
Application :
1.Projections welding is used for welding studs, nuts to plates etc.

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2.It is also used in automobile industries , ship building works anheet metal works.
3.It is also used for welding the parts of refrigerator, frills, condesers etc.

Brazing :
Process: In a brazing operation, you apply heat broadly to the base metals. The filler metal is then brought into contact with the heated parts. It is melted instantkly by the heat (greater than 427°C) in the base metals and drawn by capillary action completely through the joint.
Filler material: Alloy of Cu and Zn, Cu and Ag, Cu and Al.
Application:
1.It is used for connectors and valves requiring leak tightness.
2.Vechicle parts requiring corrosion and heat resistance.

Soldering:
Process: Soldering is a process in which metals are joined by melting a filler metal into the joint to create strong permanent bonds. Soldering may or may not have capillary attractions and is done at a temperature below 427**°C,** much lower than welding.

Filler material ; alloy of Pb and Sn.

--Flux material -- $ZnCl_2$, NH_3Cl

Application :

1.Design of electrical and electronics circuits.

2.Fabrication PCBS

Welding defects

Welding defects refer to any imperfections or irregularities that occur during the welding process, resulting in a substandard weld. The following are some common welding defects:

Porosity: Porosity is a common welding defect caused by gas trapped in the weld, which results in small holes or cavities in the weld. It can be caused by a variety of factors, such as improper shielding gas, contaminated welding wire, or inadequate welding technique.

Incomplete fusion: Incomplete fusion occurs when the weld metal fails to bond properly with the base metal or previous weld pass. It can be caused by factors such as incorrect welding technique, inadequate heat input, or poor joint preparation.

Cracking: Cracking can occur during or after the welding process and can be caused by factors such as rapid cooling, high levels of residual stress, or improper material selection.

Undercut: Undercutting occurs when the base metal at the weld joint is melted away, leaving a groove in the weld. It can be caused by excessive heat input, incorrect welding technique, or poor joint preparation.

Distortion: Distortion occurs when the welding process causes the metal to warp or deform, often resulting in misalignment or unevenness in the final product. It can be caused by factors such as high heat input, poor joint fit-up, or inadequate fixturing.

Overlapping: Overlapping occurs when two weld beads are not properly fused together, resulting in a weak and potentially hazardous weld. It can be caused by incorrect welding technique, inadequate heat input, or poor joint preparation.